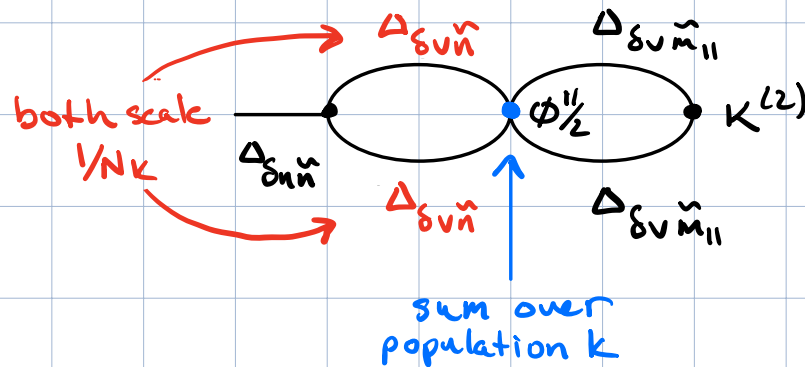
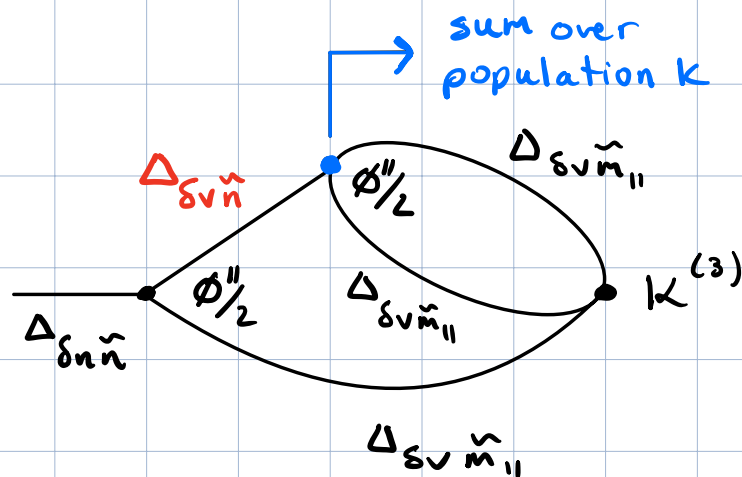


(i)



So, it will scale: $\frac{1}{N_k^2} \cdot N_k = \frac{1}{N_k}$

(ii)



So it will scale $\frac{1}{N_k} \cdot N_k = \mathcal{O}(1)$